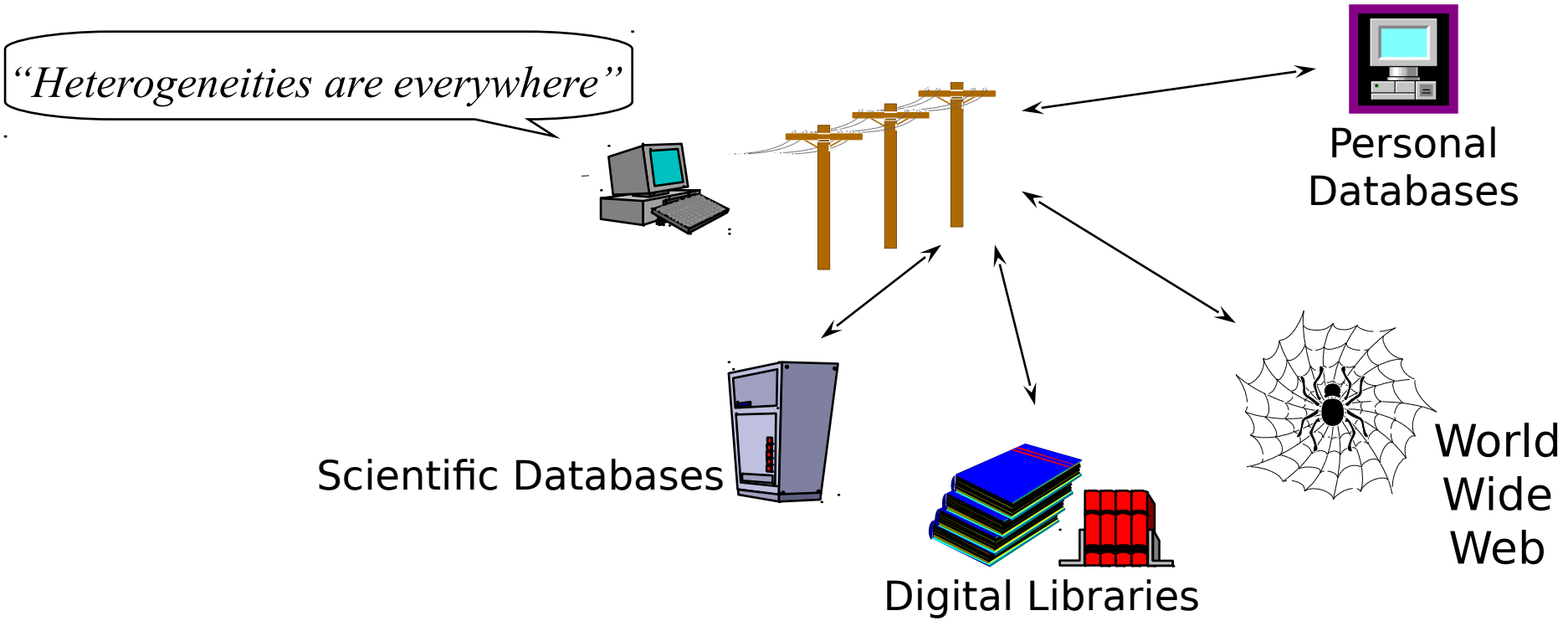
Introduction to Data Warehousing

Enrico Franconi

CS 636

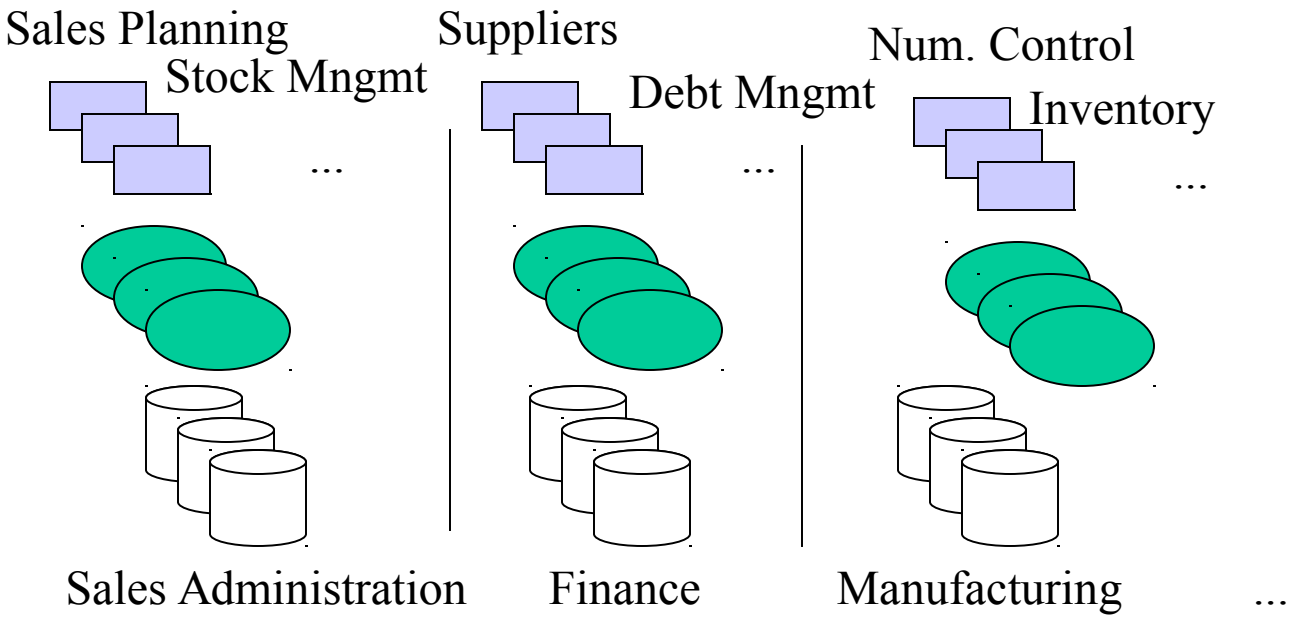
Problem: Heterogeneous Information Sources



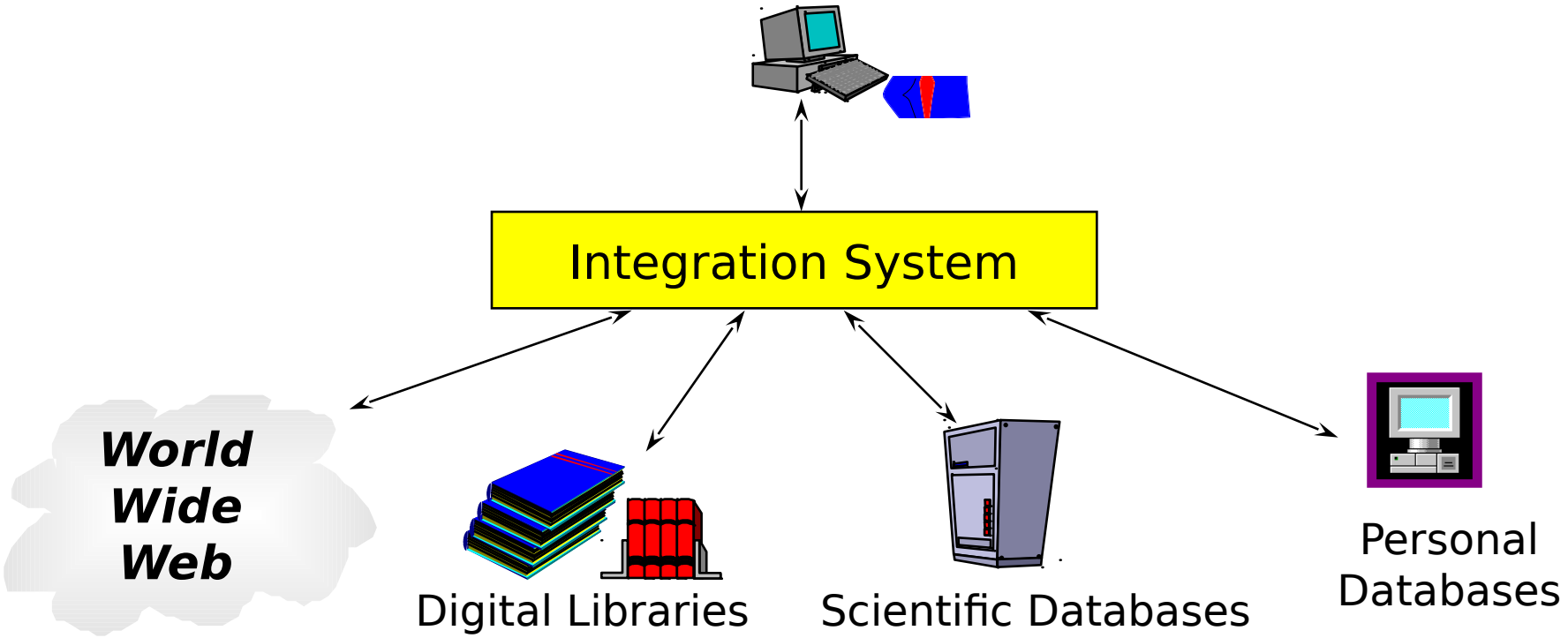
- Different interfaces
- Different data representations
- Duplicate and inconsistent information

Problem: Data Management in Large Enterprises

- Vertical fragmentation of informational systems (vertical stove pipes)
- Result of application (user)-driven development of operational systems



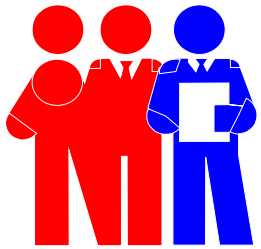
Goal: Unified Access to Data

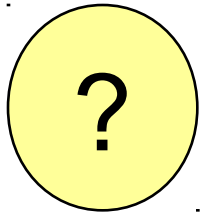


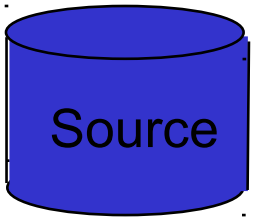
- Collects and combines information
- Provides integrated view, uniform user interface
- Supports sharing

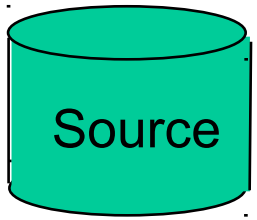
Why a Warehouse?

- Two Approaches:
  - Query-Driven (Lazy)
  - Warehouse (Eager)



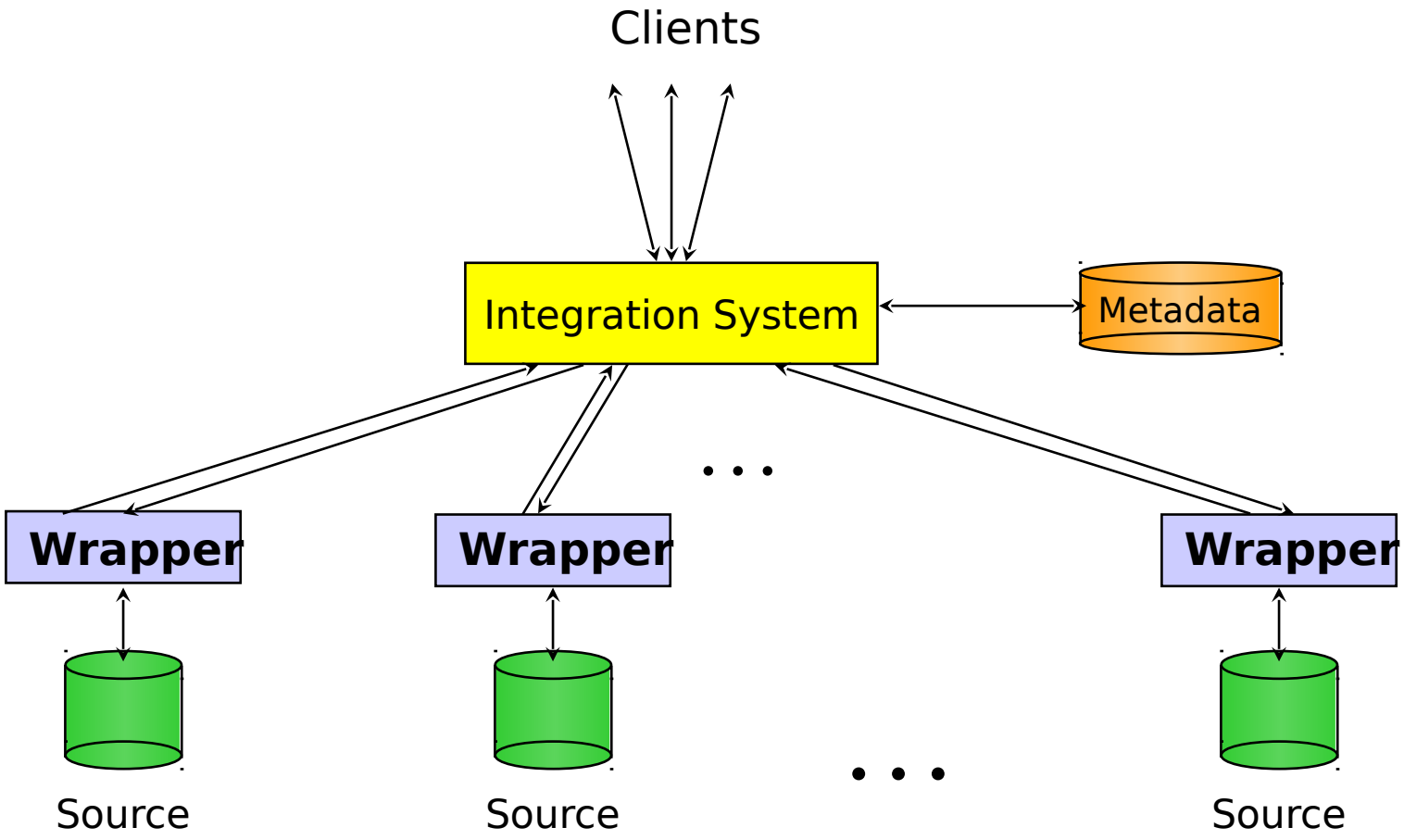






The Traditional Research Approach

- Query-driven (lazy, on-demand)

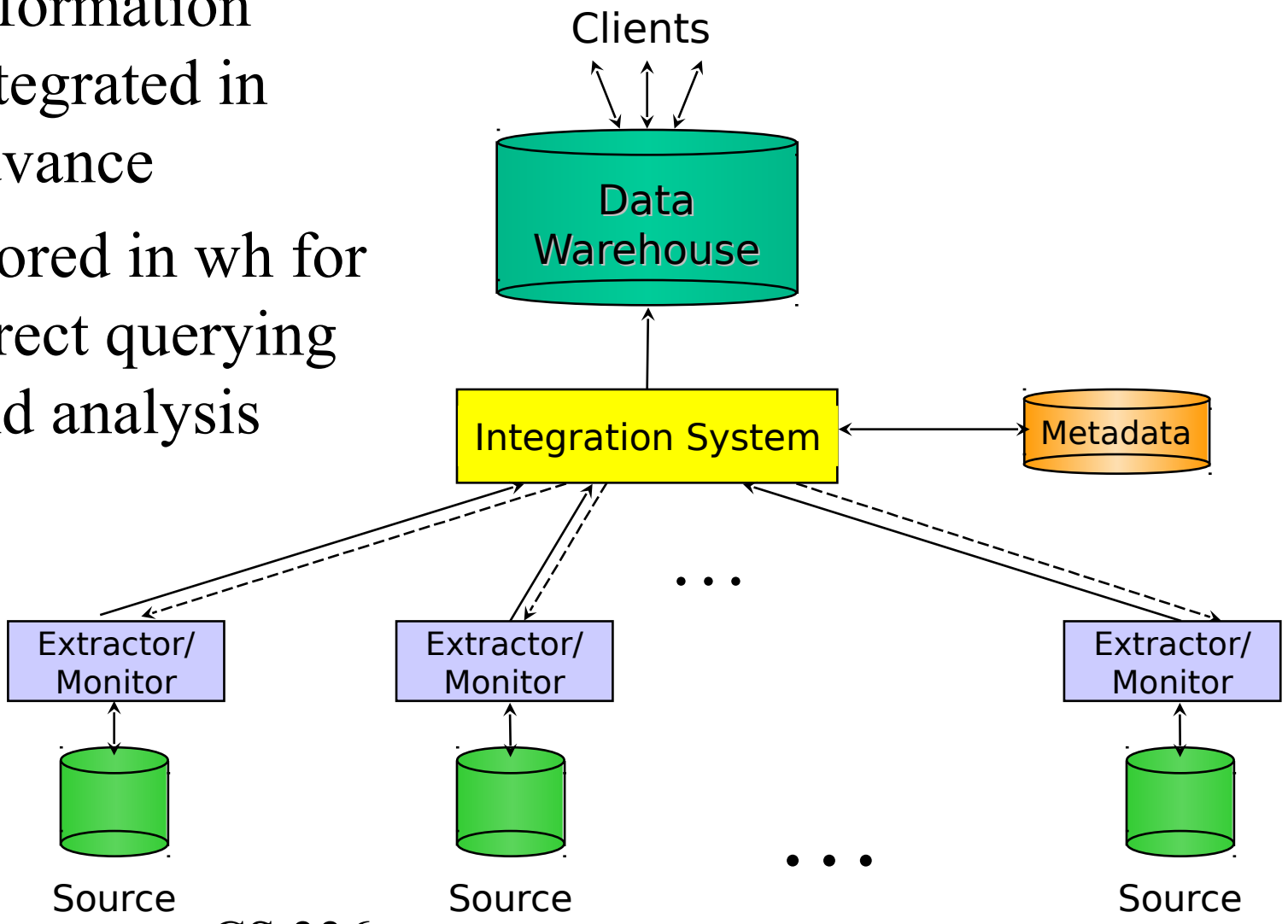


Disadvantages of Query-Driven Approach

- Delay in query processing
  - Slow or unavailable information sources
  - Complex filtering and integration
- Inefficient and potentially expensive for frequent queries
- Competes with local processing at sources
- *Hasn't* caught on in industry

The Warehousing Approach

- Information integrated in advance
- Stored in wh for direct querying and analysis



Advantages of Warehousing Approach

- High query performance
  - But not necessarily most current information
- Doesn't interfere with local processing at sources
  - Complex queries at warehouse
  - OLTP at information sources
- Information copied at warehouse
  - Can modify, annotate, summarize, restructure, etc.
  - Can store historical information
  - Security, no auditing
- *Has* caught on in industry

Not Either-Or Decision

- Query-driven approach still better for
  - Rapidly changing information
  - Rapidly changing information sources
  - Truly vast amounts of data from large numbers of sources
  - Clients with unpredictable needs

What is a Data Warehouse?

A Practitioners Viewpoint

"A data warehouse is simply a single, complete, and consistent store of data obtained from a variety of sources and made available to end users in a way they can understand and use it in a business context."

-- Barry Devlin, *IBM Consultant*

What is a Data Warehouse?

An Alternative Viewpoint

"A DW is a

- subject-oriented,
- integrated,
- time-varying,
- non-volatile

collection of data that is used primarily in organizational decision making."

-- W.H. Inmon, Building the Data Warehouse, 1992

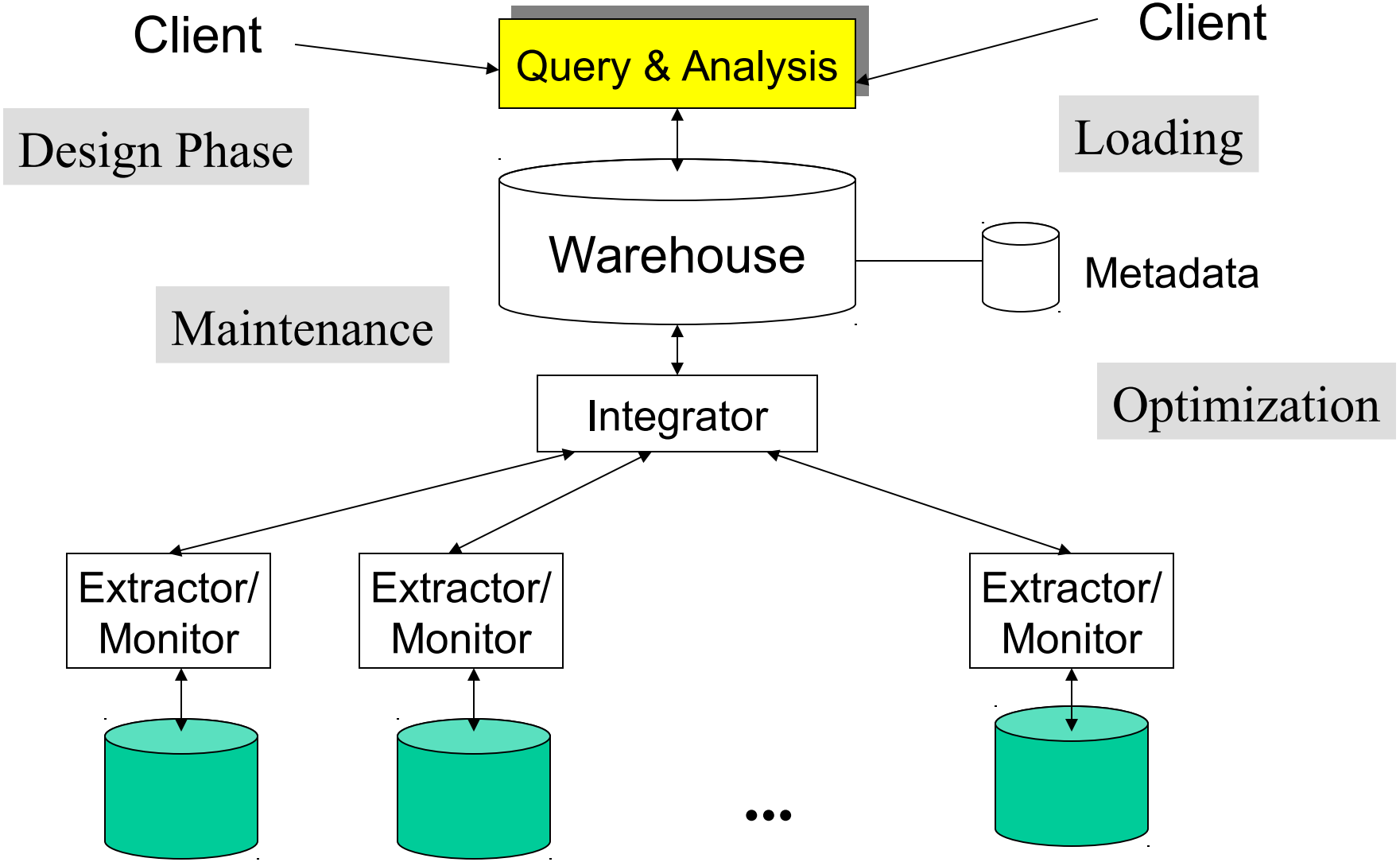
A Data Warehouse is...

- Stored collection of diverse data
  - A solution to data integration problem
  - Single repository of information
- Subject-oriented
  - Organized by subject, not by application
  - Used for analysis, data mining, etc.
- Optimized differently from transaction-oriented db
- User interface aimed at executive

… Cont'd

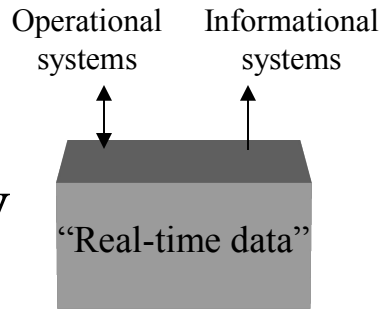
- Large volume of data (Gb, Tb)
- Non-volatile
  - Historical
  - Time attributes are important
- Updates infrequent
- May be append-only
- Examples
  - All transactions ever at Sainsbury's
  - Complete client histories at insurance firm
  - LSE financial information and portfolios

Generic Warehouse Architecture

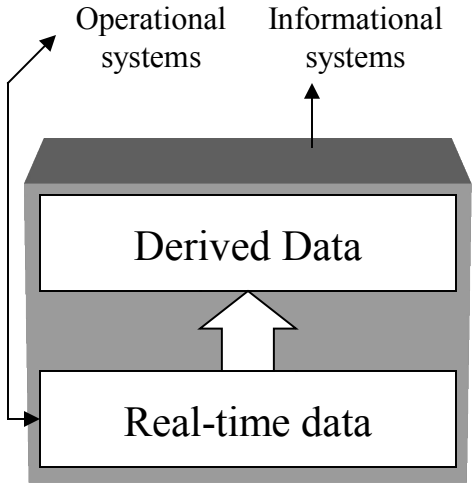


Data Warehouse Architectures: Conceptual View

- Single-layer
  - Every data element is stored once only
  - Virtual warehouse

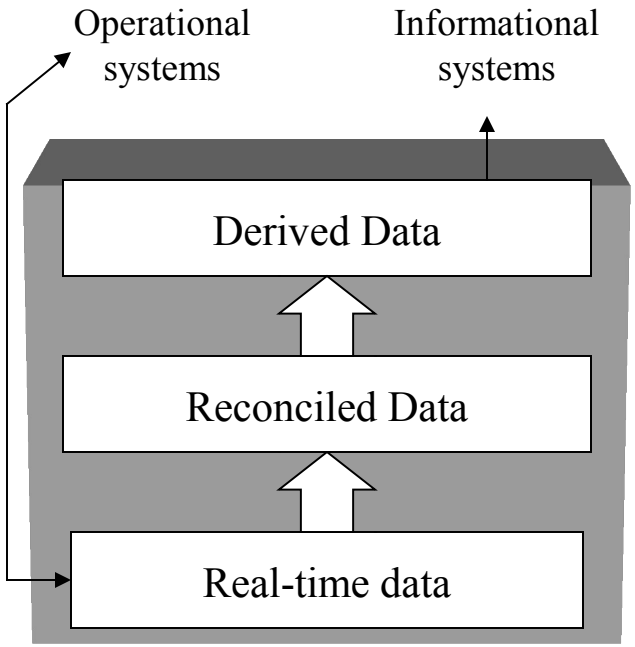


- Two-layer
  - Real-time + derived data
  - Most commonly used approach in industry today



Three-layer Architecture: Conceptual View

- Transformation of real-time data to derived data really requires two steps



View level
"Particular informational needs"

Physical Implementation of the Data Warehouse

Data Warehousing: Two Distinct Issues

(1) How to get information into warehouse

"Data warehousing"

(2) What to do with data once it's in warehouse

"Warehouse DBMS"

- Both rich research areas
- Industry has focused on (2)

Issues in Data Warehousing

- Warehouse Design
- Extraction
  - Wrappers, monitors (change detectors)
- Integration
  - Cleansing & merging
- Warehousing specification & Maintenance
- Optimizations
- Miscellaneous (e.g., evolution)

OLTP vs. OLAP

- OLTP: On Line Transaction Processing
  - Describes processing at operational sites
- OLAP: On Line Analytical Processing
  - Describes processing at warehouse

Warehouse is a Specialized DB

Standard DB (OLTP)	Warehouse (OLAP)
• Mostly updates	• Mostly reads
• Many small transactions	• Queries are long and complex
• Mb - Gb of data	• Gb - Tb of data
• Current snapshot	• History
• Index/hash on p.k.	• Lots of scans
• Raw data	• Summarized, reconciled data
• Thousands of users (e.g., clerical users)	• Hundreds of users (e.g., decision-makers, analysts)